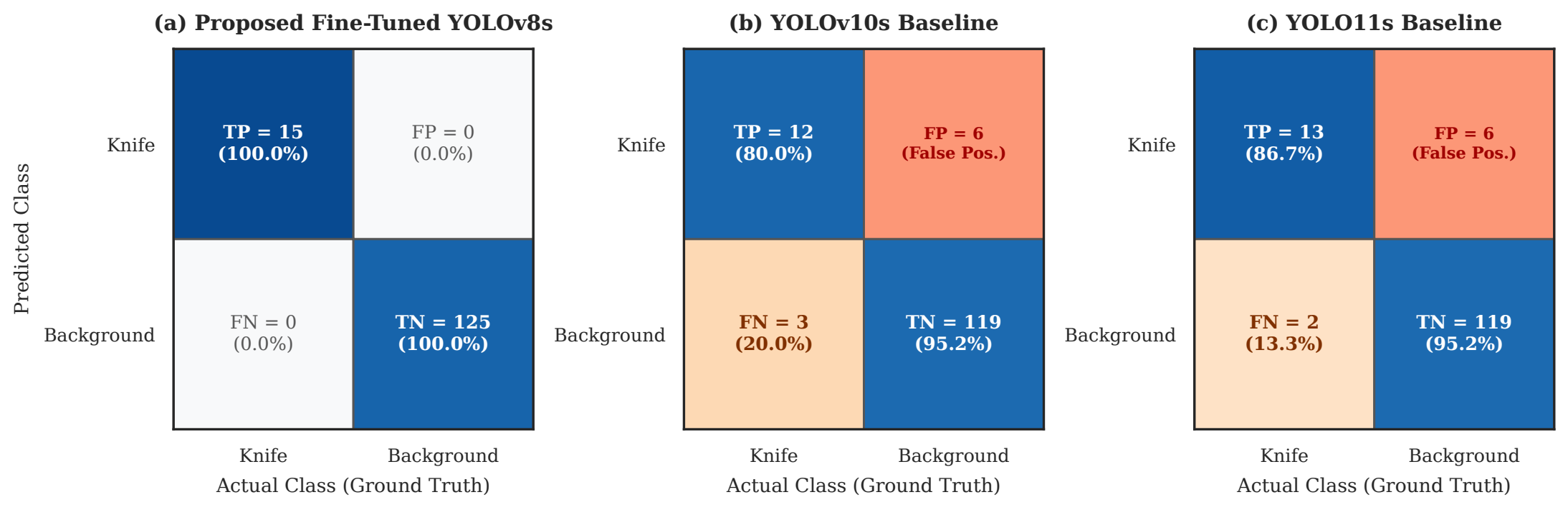
(a) Proposed Fine-Tuned YOLOv8s
(b) YOLOv10s Baseline
(c) YOLO11s Baseline
Predicted Class
Knife
Background
TP = 15
(100.0%)
FP = 0
(0.0%)
FN = 0
(0.0%)
TN = 125
(100.0%)
TP = 12
(80.0%)
FP = 6
(False Pos.)
FN = 3
(20.0%)
TN = 119
(95.2%)
TP = 13
(86.7%)
FP = 6
(False Pos.)
FN = 2
(13.3%)
TN = 119
(95.2%)
Knife
Background
Actual Class (Ground Truth)